

1. COMPUTER NETWORKS AND CLOUD COMPUTING

(10%)

Q1. Data communication involves the transfer of data between two devices via some form of:

- A) Software
- B) Transmission medium ☒
- C) Operating system
- D) Database

Q2. Which transmission mode allows data to flow in both directions simultaneously?

- A) Simplex
- B) Half-duplex
- C) Full-duplex ☒
- D) Multiplex

Q3. What is the maximum number of layers in the TCP/IP model?

- A) 7
- B) 5 ☒
- C) 4
- D) 6

Q4. Which type of network covers a city-sized area?

- A) LAN
- B) WAN
- C) MAN ☒
- D) PAN

Q5. Which Data Link Layer protocol is used in Ethernet LANs?

- A) PPP
- B) CSMA/CD ☒
- C) HDLC
- D) Token Ring only

Q6. What is the size of an IPv4 address?

- A) 64 bits
- B) 128 bits
- C) 32 bits ☒
- D) 48 bits

Q7. Which Network Layer protocol is used to map IP addresses to MAC addresses?

- A) RARP
- B) ARP ☒
- C) ICMP
- D) IGMP

Q8. Which Transport Layer feature ensures flow control using a sliding window?

- A) Multiplexing
- B) TCP sliding window protocol ☒
- C) UDP broadcasting
- D) IP fragmentation

Q9. The port number for HTTP is:

- A) 21
- B) 25
- C) 80 ☒
- D) 443

Q10. Which Application Layer protocol is used for sending emails?

- A) FTP
- B) HTTP
- C) SMTP ☒
- D) SNMP

Q11. In Wi-Fi, which access method is used to avoid collisions?

- A) CSMA/CD
- B) CSMA/CA ☒
- C) Token passing
- D) Polling

Q12. Which cloud deployment model combines both public and private cloud features?

- A) Community Cloud
- B) Hybrid Cloud ☒
- C) Multi-Cloud
- D) Virtual Cloud

Q13. PaaS stands for:

- A) Protocol as a Service
- B) Platform as a Service ☒
- C) Processing as a Service
- D) Packet as a Service

Q14. A firewall operates primarily to:

- A) Speed up internet
- B) Filter incoming and outgoing network traffic based on security rules ☒
- C) Store data
- D) Assign IP addresses

Q15. SDN stands for:

- A) Simple Data Network
- B) Software Defined Networking ☒
- C) Secure Digital Network
- D) Standard Distribution Node

2. PROGRAMMING (C++/Java/Python) (10%)

Q16. Which of the following is NOT a valid variable name in Python?

- A) _myVar
- B) my_var
- C) 2ndVar ☒
- D) myVar2

Q17. What is the result of `bool("False")` in Python?

- A) False
- B) True ☒
- C) Error
- D) None

Q18. In C++, which operator is used for scope resolution?

- A) .
- B) ->
- C) :: ☒
- D) #

Q19. What does the `break` statement do inside a loop?

- A) Skips the current iteration
- B) Exits the loop immediately ☒
- C) Restarts the loop
- D) Pauses the loop

Q20. The `continue` statement in a loop:

- A) Terminates the loop
- B) Skips the remaining code in the current iteration and moves to the next ☒

C) Returns a value

D) Creates a new loop

Q21. What is function overloading?

A) Calling a function multiple times

B) Multiple functions with the same name but different parameters ☒

C) A function calling itself

D) A function with no return type

Q22. In Java, which method is the entry point of a program?

A) start()

B) main() ☒

C) run()

D) init()

Q23. What is the output of len("Hello World") in Python?

A) 10

B) 11 ☒

C) 12

D) 9

Q24. Which of the following is used to read a file in Python?

A) open("file.txt", "w")

B) open("file.txt", "r") ☒

C) open("file.txt", "a")

D) open("file.txt", "d")

Q25. In Python, a dictionary stores data as:

A) Ordered sequence

B) Key-value pairs ☒

C) Linked nodes

D) Binary tree

Q26. Which OOP concept hides internal details and shows only functionality?

A) Polymorphism

B) Inheritance

C) Abstraction ☒

D) Composition

Q27. What is a constructor in OOP?

A) A method that destroys objects

B) A special method called automatically when an object is created ☒

C) A static method

D) A private function

Q28. In C++, which keyword is used to free dynamically allocated memory?

- A) free
- B) delete ☒
- C) remove
- D) destroy

Q29. What is a segmentation fault?

- A) A syntax error
- B) Accessing memory that the program is not allowed to access ☒
- C) A logical error
- D) A compilation warning

Q30. In Python, which keyword is used to define a generator function?

- A) return
- B) yield ☒
- C) generate
- D) produce

Q31. What is the purpose of pip in Python?

- A) Compiler
- B) Package installer ☒
- C) Debugger
- D) IDE

3. DATA STRUCTURES & ALGORITHMS (10%)

Q32. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ ☒
- D) $O(n^2)$

Q33. In a singly linked list, each node contains:

- A) Data only
- B) Data and pointer to the next node ☒
- C) Data and pointer to previous node
- D) Two data fields

Q34. Which type of linked list has the last node pointing to the first node?

- A) Singly linked list
- B) Doubly linked list

C) Circular linked list ☒

D) Skip list

Q35. What is the maximum number of children a node can have in a binary tree?

A) 1

B) 2 ☒

C) 3

D) Unlimited

Q36. Which tree traversal visits: Left → Root → Right?

A) Pre-order

B) In-order ☒

C) Post-order

D) Level-order

Q37. In a min-heap, the root element is:

A) The largest element

B) The smallest element ☒

C) The median element

D) Random

Q38. Which searching algorithm requires the data to be sorted?

A) Linear search

B) Binary search ☒

C) Sequential search

D) Depth-first search

Q39. What is the best-case time complexity of Bubble Sort?

A) $O(n^2)$

B) $O(n \log n)$

C) $O(n)$ ☒

D) $O(1)$

Q40. Which sorting algorithm has the best average-case time complexity?

A) Bubble Sort — $O(n^2)$

B) Selection Sort — $O(n^2)$

C) Merge Sort — $O(n \log n)$ ☒

D) Insertion Sort — $O(n^2)$

Q41. Open addressing and chaining are techniques used to resolve:

A) Stack overflow

B) Hash collisions ☒

C) Queue underflow

D) Tree imbalance

Q42. An AVL tree is a:

A) Binary tree with no balancing

B) Self-balancing binary search tree ☒

C) Multi-way search tree

D) Graph

Q43. Which graph traversal algorithm uses a stack (or recursion)?

A) BFS

B) DFS ☒

C) Dijkstra's

D) Prim's

Q44. Prim's and Kruskal's algorithms are used to find:

A) Shortest path

B) Minimum spanning tree ☒

C) Maximum flow

D) Topological order

Q45. Which algorithm design technique makes the best local choice at each step hoping to find global optimum?

A) Dynamic Programming

B) Divide and Conquer

C) Greedy ☒

D) Backtracking

Q46. The Fibonacci sequence computed using dynamic programming has time complexity:

A) $O(2^n)$

B) $O(n)$ ☒

C) $O(n^2)$

D) $O(n \log n)$

Q47. A trie data structure is primarily used for:

A) Numerical sorting

B) Efficient string searching and prefix matching ☒

C) Graph traversal

D) Matrix multiplication

Q48. Amortized analysis is used to:

A) Find the worst case of a single operation

B) Calculate average cost per operation over a sequence of operations ☒

- C) Find best case only
- D) Compare two algorithms

4. OPERATING SYSTEMS (5%)

Q49. Which type of operating system allows multiple users to use the system simultaneously?

- A) Batch OS
- B) Time-sharing OS ☒
- C) Real-time OS
- D) Embedded OS

Q50. The component of the OS that manages hardware directly is called the:

- A) Shell
- B) Kernel ☒
- C) GUI
- D) Utility

Q51. In process management, the state of a process that is waiting for I/O is:

- A) Running
- B) Ready
- C) Blocked/Waiting ☒
- D) Terminated

Q52. Which CPU scheduling algorithm assigns a fixed time quantum to each process?

- A) FCFS
- B) SJF
- C) Round Robin ☒
- D) Priority

Q53. Context switching refers to:

- A) Switching between users
- B) Saving and restoring the state of a process so another process can run ☒
- C) Switching between hard drives
- D) Changing the OS

Q54. A semaphore is used for:

- A) Memory allocation
- B) Process synchronization ☒

C) File deletion

D) Network routing

Q55. Which deadlock handling strategy simply ignores the deadlock problem?

A) Prevention

B) Avoidance

C) Detection and Recovery

D) Ostrich Algorithm ☒

Q56. Virtual memory allows:

A) Programs larger than physical memory to execute ☒

B) Faster CPU processing

C) More RAM to be installed

D) Files to be compressed

Q57. Which page replacement algorithm replaces the page that will not be used for the longest time in future?

A) FIFO

B) LRU

C) Optimal (OPT) ☒

D) Clock

Q58. Which file allocation method stores files in contiguous blocks?

A) Linked allocation

B) Indexed allocation

C) Contiguous allocation ☒

D) Hashed allocation

Q59. RAID stands for:

A) Random Access Integrated Disk

B) Redundant Array of Independent Disks ☒

C) Rapid Access Input Device

D) Reliable Array of Internal Drives

Q60. In OS, DMA stands for:

A) Direct Memory Access ☒

B) Dynamic Memory Allocation

C) Disk Management Application

D) Data Management Architecture

Q61. Access Control Lists (ACLs) are used for:

A) CPU scheduling

B) Defining permissions for resources ☒

- C) Memory management
- D) Process creation

5. SOFTWARE ENGINEERING (10%)

Q62. The Spiral Model primarily focuses on:

- A) Sequential development
- B) Risk analysis ☒
- C) Code reuse
- D) Fast deployment

Q63. In Scrum (Agile), the product owner is responsible for:

- A) Writing all the code
- B) Managing and prioritizing the product backlog ☒
- C) Testing the software
- D) Deploying the application

Q64. Functional requirements describe:

- A) How fast the system should respond
- B) What the system should do ☒
- C) System security needs
- D) Hardware specifications

Q65. Non-functional requirements include:

- A) User login functionality
- B) Performance, security, and usability constraints ☒
- C) Database queries
- D) Report generation

Q66. A Gantt chart is used in software project management to:

- A) Write code
- B) Visualize project schedule and task timelines ☒
- C) Test software
- D) Design databases

Q67. Coupling in software design refers to:

- A) How closely classes within a module are related
- B) The degree of interdependence between modules ☒
- C) The size of the code
- D) The number of test cases

Q68. High cohesion and low coupling is considered:

- A) Bad design
- B) Good design ☒
- C) Irrelevant
- D) Only applicable to databases

Q69. Which software architecture style uses layers like presentation, business logic, and data access?

- A) Client-Server
- B) Microservices
- C) Layered (N-tier) Architecture ☒
- D) Pipe and Filter

Q70. A wireframe in UI design is:

- A) The final product
- B) A low-fidelity visual representation of a user interface ☒
- C) A programming language
- D) A database schema

Q71. Integration testing is performed:

- A) Before unit testing
- B) After unit testing, to test combined modules ☒
- C) Only after deployment
- D) During requirements gathering

Q72. Regression testing ensures:

- A) New features work
- B) Previously working features still work after changes ☒
- C) The system meets performance requirements
- D) Code is properly documented

Q73. What is refactoring?

- A) Adding new features
- B) Restructuring existing code without changing its external behavior ☒
- C) Deleting old code
- D) Writing documentation

Q74. Lines of Code (LOC) is an example of:

- A) A software testing method
- B) A software metric ☒
- C) A design pattern
- D) An algorithm

Q75. In software configuration management, a "baseline" is:

- A) The first line of code
- B) A formally reviewed and agreed-upon specification that serves as the basis for further development ☒
- C) A backup copy
- D) A test case

Q76. Risk mitigation in software engineering means:

- A) Ignoring risks
- B) Taking steps to reduce the probability or impact of identified risks ☒
- C) Adding more features
- D) Increasing the budget

Q77. Secure coding practices include:

- A) Ignoring user input validation
- B) Input validation, proper error handling, and avoiding hard-coded credentials ☒
- C) Using plain text passwords
- D) Disabling firewalls

6. WEB DEVELOPMENT (10%)

Q78. Which protocol ensures secure communication over the web using encryption?

- A) HTTP
- B) FTP
- C) HTTPS ☒
- D) SMTP

Q79. The <div> tag in HTML is used for:

- A) Creating links
- B) Defining a division or section in a document ☒
- C) Inserting images
- D) Creating tables

Q80. Which HTML5 element is used for navigation links?

- A) <header>
- B) <nav> ☒
- C) <section>
- D) <footer>

Q81. Which CSS property is used to create space inside an element's border?

- A) margin
- B) padding ☒
- C) border
- D) spacing

Q82. CSS Flexbox is used for:

- A) Database queries
- B) One-dimensional layout (row or column) ☒
- C) Server-side processing
- D) Image editing

Q83. In CSS, @media queries are used for:

- A) Importing fonts
- B) Making responsive designs that adapt to different screen sizes ☒
- C) Connecting to databases
- D) Adding animations only

Q84. Which JavaScript method is used to select an HTML element by its ID?

- A) querySelector()
- B) getElementById() ☒
- C) getElementByClass()
- D) selectById()

Q85. What is the DOM in web development?

- A) Data Object Model
- B) Document Object Model — a programming interface for HTML documents ☒
- C) Digital Output Method
- D) Database Object Manager

Q86. In JavaScript, async/await is used for:

- A) Styling web pages
- B) Handling asynchronous operations ☒
- C) Database design
- D) Creating HTML elements

Q87. Which frontend framework is developed by Google?

- A) React
- B) Angular ☒
- C) Vue.js
- D) Svelte

Q88. Express.js is a framework for:

- A) Frontend development

B) Backend development with Node.js ☒

C) CSS styling

D) Database management

Q89. In server-side programming, PHP originally stood for:

A) Private Home Page

B) Personal Home Page (now PHP: Hypertext Preprocessor) ☒

C) Public Host Protocol

D) Program Handling Process

Q90. An ORM (Object-Relational Mapping) is used to:

A) Design user interfaces

B) Map database tables to objects in programming languages ☒

C) Encrypt data

D) Compress files

Q91. CORS stands for:

A) Cross-Origin Request Sharing

B) Cross-Origin Resource Sharing ☒

C) Client-Origin Resource System

D) Central Origin Response Service

Q92. Which of the following is a common technique for web performance optimization?

A) Adding more images

B) Minifying CSS and JavaScript files ☒

C) Using larger fonts

D) Disabling caching

Q93. Selenium is used for:

A) Backend programming

B) Automated web testing ☒

C) Database management

D) CSS styling

Q94. Which service is commonly used for continuous deployment?

A) Photoshop

B) GitHub Actions / Jenkins ☒

C) Microsoft Word

D) VLC Player

Q95. A REST API typically uses which data format for responses?

A) XML only

B) JSON ☒

C) CSV

D) Binary

Q96. OAuth is used for:

A) Database management

B) Authorization and access delegation ☒

C) CSS styling

D) HTML rendering

Q97. Progressive Web Apps (PWAs) are:

A) Desktop applications only

B) Web applications that can work offline and provide app-like experience ☒

C) Only mobile apps

D) Database tools

7. AI / MACHINE LEARNING AND DATA ANALYTICS

(10%)

Q98. The Turing Test is used to evaluate:

A) Network speed

B) A machine's ability to exhibit intelligent behavior indistinguishable from a human



C) Database performance

D) Sorting algorithms

Q99. Gradient descent is an optimization algorithm used to:

A) Sort data

B) Minimize the cost/loss function in machine learning ☒

C) Encrypt data

D) Compress images

Q100. In linear algebra, a matrix with the same number of rows and columns is called:

A) Rectangular matrix

B) Square matrix ☒

C) Identity matrix

D) Null matrix

Q101. Bayes' theorem is used in:

- A) Sorting algorithms
- B) Probability-based classification (e.g., Naive Bayes) ☒
- C) Graph traversal
- D) File management

Q102. NumPy in Python is primarily used for:

- A) Web development
- B) Numerical computing and array operations ☒
- C) GUI development
- D) Database management

Q103. Which Python library is used for data visualization?

- A) Pandas
- B) Matplotlib ☒
- C) Scikit-learn
- D) Flask

Q104. Data preprocessing includes:

- A) Only data collection
- B) Handling missing values, normalization, encoding categorical data ☒
- C) Model deployment
- D) Writing reports

Q105. Feature scaling ensures that:

- A) All features are deleted
- B) Features are on a similar scale so no feature dominates ☒
- C) Data is encrypted
- D) Models run faster without data

Q106. A box plot is used to visualize:

- A) Only mean values
- B) Distribution of data showing median, quartiles, and outliers ☒
- C) Network topology
- D) File structure

Q107. In supervised learning, linear regression is used for:

- A) Classification
- B) Predicting continuous numerical values ☒
- C) Clustering
- D) Dimensionality reduction

Q108. Logistic regression is used for:

- A) Regression problems

B) Binary classification problems ☒

C) Clustering

D) Image processing

Q109. Support Vector Machine (SVM) finds:

A) The average of all data points

B) The optimal hyperplane that separates classes ☒

C) The nearest neighbor

D) The shortest path

Q110. In unsupervised learning, which technique reduces the number of features?

A) Classification

B) PCA (Principal Component Analysis) ☒

C) Linear Regression

D) Cross-validation

Q111. Cross-validation is used to:

A) Train the final model

B) Evaluate model performance by splitting data into multiple folds ☒

C) Deploy the model

D) Collect more data

Q112. In a neural network, an activation function:

A) Stores weights

B) Introduces non-linearity to the network ☒

C) Loads data

D) Saves the model

Q113. Transfer learning means:

A) Moving data between databases

B) Using a pre-trained model on a new but related task ☒

C) Transferring files

D) Copying code

Q114. Recall in classification measures:

A) How many predicted positives are actually positive

B) How many actual positives are correctly identified ☒

C) Overall accuracy

D) Model speed

Q115. Which technique involves creating multiple models and combining their predictions?

A) Single model learning

- B) Ensemble methods (Bagging, Boosting) ☒
- C) Transfer learning
- D) Unsupervised learning

8. CYBER SECURITY (5%)

Q116. A DDoS attack aims to:

- A) Steal passwords
- B) Overwhelm a system with traffic making it unavailable ☒
- C) Encrypt files for ransom
- D) Modify database records

Q117. RSA is an example of:

- A) Symmetric encryption
- B) Asymmetric encryption ☒
- C) Hashing algorithm
- D) Compression technique

Q118. A digital certificate is issued by:

- A) An ISP
- B) A Certificate Authority (CA) ☒
- C) The user themselves
- D) The operating system

Q119. Which type of malware locks user files and demands payment?

- A) Virus
- B) Worm
- C) Ransomware ☒
- D) Adware

Q120. An Intrusion Detection System (IDS) is used to:

- A) Block all traffic
- B) Monitor network traffic for suspicious activity ☒
- C) Create backups
- D) Assign IP addresses

Q121. The principle of least privilege means:

- A) Giving everyone admin access
- B) Users should only have the minimum access necessary to perform their tasks ☒

C) Removing all permissions

D) Sharing passwords

Q122. What is a man-in-the-middle (MITM) attack?

A) An attacker intercepts communication between two parties ☒

B) An attacker creates a virus

C) An attacker steals hardware

D) An attacker shuts down a server

Q123. Chain of custody in digital forensics refers to:

A) Installing antivirus

B) The documented process of handling digital evidence to maintain its integrity ☒

C) Writing code

D) Creating backups

9. DATABASES (10%)

Q124. An ER diagram is used to:

A) Write SQL queries

B) Model the conceptual design of a database showing entities and relationships



C) Execute programs

D) Design web pages

Q125. A foreign key is:

A) A unique identifier for each row

B) An attribute in one table that references the primary key of another table ☒

C) A key used for encryption

D) An index

Q126. Which SQL command is used to modify existing data in a table?

A) INSERT

B) UPDATE ☒

C) ALTER

D) SELECT

Q127. The DDL command used to remove a table from the database is:

A) DELETE

B) REMOVE

C) DROP ☒

D) TRUNCATE

Q128. First Normal Form (1NF) requires:

A) No transitive dependencies

B) Elimination of repeating groups — each column must contain atomic values ☒

C) No partial dependencies

D) All attributes depend on the full key

Q129. Second Normal Form (2NF) requires:

A) Eliminating transitive dependencies

B) Being in 1NF and eliminating partial dependencies ☒

C) Having no multi-valued attributes

D) All keys are primary

Q130. A view in SQL is:

A) A physical table

B) A virtual table based on the result of a SQL query ☒

C) A backup copy

D) An index

Q131. Which SQL clause is used to sort the result set?

A) GROUP BY

B) HAVING

C) ORDER BY ☒

D) WHERE

Q132. A stored procedure is:

A) A single SQL statement

B) A precompiled collection of SQL statements stored in the database ☒

C) A type of table

D) A backup file

Q133. In transaction management, "Atomicity" means:

A) Transactions can be split

B) A transaction is either fully completed or fully rolled back ☒

C) Data must be consistent

D) Changes are permanent

Q134. Deadlock in databases occurs when:

A) A query takes too long

B) Two or more transactions are waiting for each other to release locks ☒

C) A table is deleted

D) An index is created

Q135. Which of the following is a NoSQL database type?

A) Relational

B) Document-based (e.g., MongoDB) ☒

C) Spreadsheet

D) Sequential

Q136. In data warehousing, ETL stands for:

A) Execute, Transform, Load

B) Extract, Transform, Load ☒

C) Enter, Test, Log

D) Evaluate, Transfer, Link

Q137. OLAP stands for:

A) Online Analytical Processing ☒

B) Online Application Protocol

C) Offline Logical Analysis Processing

D) Open Link Access Protocol

Q138. Which SQL aggregate function returns the total number of rows?

A) SUM()

B) AVG()

C) COUNT() ☒

D) MAX()

10. PROBLEM SOLVING AND ANALYTICAL SKILLS (20%)

Q139. What is the next number: 1, 1, 2, 3, 5, 8, ?

A) 11

B) 12

C) 13 ☒

D) 15

Explanation: Fibonacci — each number is the sum of the two preceding ones.

$5+8=13$

Q140. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?

- A) 100 minutes
- B) 50 minutes
- C) 5 minutes ☒
- D) 1 minute

Explanation: Each machine takes 5 minutes to make one widget. 100 machines each make 1 widget in 5 minutes.

Q141. Which of the following represents $O(n \log n)$ complexity?

- A) Accessing an array element
- B) Linear search
- C) Merge Sort ☒
- D) Bubble Sort

Q142. A pseudocode is:

- A) Actual executable code
- B) An informal high-level description of a program's logic ☒
- C) A compiler
- D) A type of database

Q143. What is the decimal equivalent of binary 11010?

- A) 24
- B) 26 ☒
- C) 28
- D) 22

Explanation: $16+8+0+2+0 = 26$

Q144. Which of the following is the hexadecimal equivalent of decimal 255?

- A) FF ☒
- B) FE
- C) EF
- D) 1F

Q145. If $x = 7$ and $y = 3$, what is $x \& y$ (bitwise AND)?

- A) 7
- B) 3 ☒
- C) 4
- D) 1

Explanation: $111 \& 011 = 011 = 3$

Q146. A recursive solution must have:

- A) Only recursive calls

B) A base case and a recursive case ☒

C) No return statement

D) Infinite loops

Q147. What type of error is: dividing by zero at runtime?

A) Syntax error

B) Logical error

C) Runtime error/exception ☒

D) Compilation error

Q148. When solving a problem, which approach builds from the simplest sub-solutions upward?

A) Top-down

B) Bottom-up ☒

C) Random

D) Brute force

Q149. What is the output of this pseudocode?

text

x = 0

for i = 1 to 4:

 x = x + i

print x

A) 4

B) 6

C) 10 ☒

D) 15

Explanation: $1+2+3+4 = 10$

Q150. A parking lot has 3 levels. Level 1 has 50 spots, Level 2 has 40 spots, Level 3 has 30 spots. If 65% of total spots are filled, how many cars are parked?

A) 72

B) 78 ☒

C) 80

D) 85

Explanation: Total = 120, 65% of 120 = 78

Q151. In analytical reasoning: All programmers use logic. Some students are programmers. Therefore:

A) All students use logic

B) Some students use logic ☒

- C) No students use logic
- D) All programmers are students

Q152. Which data representation shows the relationship between two variables?

- A) Pie chart
- B) Scatter plot ☒
- C) Bar chart
- D) Histogram

Q153. Space complexity of an algorithm refers to:

- A) The physical size of the code file
- B) The amount of memory an algorithm uses relative to input size ☒
- C) Disk space needed
- D) Network bandwidth

Q154. What comes next: 3, 9, 27, 81, ?

- A) 162
- B) 243 ☒
- C) 180
- D) 216

Explanation: Each number is multiplied by 3. $81 \times 3 = 243$

Q155. When solving a complex system design problem, which approach is most effective?

- A) Writing code immediately
- B) Breaking the problem into components, designing interfaces, then implementing ☒
- C) Guessing the solution
- D) Copying existing code

Q156. If a function has time complexity $T(n) = 2T(n/2) + n$, by the Master Theorem it is:

- A) $O(n)$
- B) $O(n \log n)$ ☒
- C) $O(n^2)$
- D) $O(\log n)$

Q157. A truth table is used to determine:

- A) The speed of a program
- B) The output of logical expressions for all input combinations ☒
- C) Memory usage
- D) Network performance

Q158. Pattern recognition is the ability to:

A) Write code

B) Identify similarities, trends, or regularities in data or problems 

C) Delete files

D) Create databases

ANSWER KEY SUMMARY

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	41	B	81	B	121	B
2	C	42	B	82	B	122	A
3	B	43	B	83	B	123	B
4	C	44	B	84	B	124	B
5	B	45	C	85	B	125	B
6	C	46	B	86	B	126	B
7	B	47	B	87	B	127	C
8	B	48	B	88	B	128	B
9	C	49	B	89	B	129	B
10	C	50	B	90	B	130	B
11	B	51	C	91	B	131	C
12	B	52	C	92	B	132	B
13	B	53	B	93	B	133	B
14	B	54	B	94	B	134	B
15	B	55	D	95	B	135	B
16	C	56	A	96	B	136	B
17	B	57	C	97	B	137	A
18	C	58	C	98	B	138	C
19	B	59	B	99	B	139	C
20	B	60	A	100	B	140	C
21	B	61	B	101	B	141	C
22	B	62	B	102	B	142	B
23	B	63	B	103	B	143	B

24	B	64	B	104	B	144	A
25	B	65	B	105	B	145	B
26	C	66	B	106	B	146	B
27	B	67	B	107	B	147	C
28	B	68	B	108	B	148	B
29	B	69	C	109	B	149	C
30	B	70	B	110	B	150	B
31	B	71	B	111	B	151	B
32	C	72	B	112	B	152	B
33	B	73	B	113	B	153	B
34	C	74	B	114	B	154	B
35	B	75	B	115	B	155	B
36	B	76	B	116	B	156	B
37	B	77	B	117	B	157	B
38	B	78	C	118	B	158	B
39	C	79	B	119	C		
40	C	80	B	120	B		